

Simple Graph

Graph Theory Explorer — Study Notes

Definition

A simple graph is a graph that has no loops (edges connecting a vertex to itself) and no multiple edges (more than one edge between the same pair of vertices). Every pair of vertices is connected by at most one edge.

Key Properties

- No self-loops: an edge cannot start and end at the same vertex.
- No parallel/multiple edges: at most one edge exists between any two distinct vertices.
- For n vertices, the maximum possible number of edges in a simple undirected graph is $n(n-1)/2$.
- Most foundational graph theory results (Euler's formula, graph coloring, etc.) are stated for simple graphs.
- A simple graph can be either directed or undirected, as long as the no-loop / no-multi-edge rules hold.

Example

Let $V = \{A, B, C\}$ with edges $E = \{(A,B), (B,C), (A,C)\}$. Each pair of vertices is joined by exactly one edge, and no vertex connects to itself. This forms a triangle — a simple graph with 3 vertices and 3 edges.

Why It Matters

Simple graphs are the default assumption in most graph theory contexts. Understanding what makes a graph 'simple' helps distinguish it from multigraphs (which allow repeated edges) and pseudographs (which allow loops), both of which require different mathematical treatment.